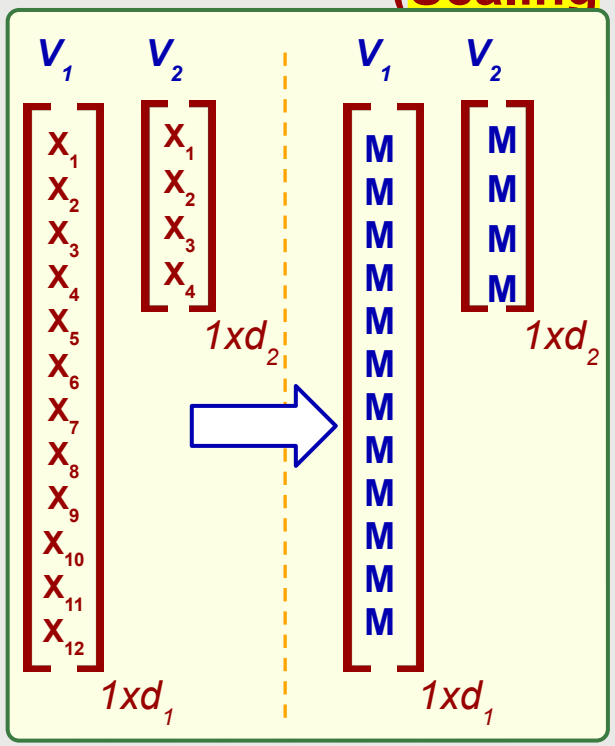
Self Attention Mechanism
(Scaling based on Square Root of Embedding Vector Dimension)

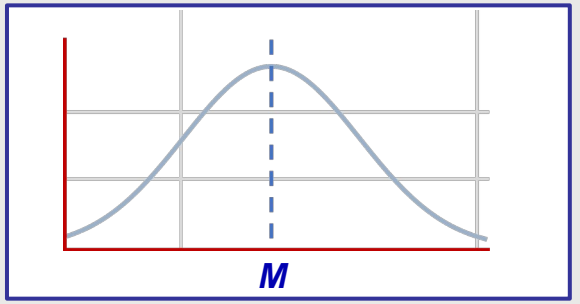


Let Average Values of elements of V_1 = **M**

Let Average Values of elements of V_2 = **M**

Vector Length of V_1 (all Ms) = $V_1^T V_1$ = M $\sqrt{d_1}$

Vector Length of V_2 (all Ms) = $V_2^T V_2$ = M $\sqrt{d_2}$



$\sqrt{d}$ ***is a good choice for scaling***

$\sqrt[4]{d}$ ***Some times is used***

SoftMax $[w'_{ij}]$

SoftMax $\left[\frac{w'_{ij}}{\sqrt{d}}\right]$

d=100, $\sqrt{d}$ **=10**

Remember:
"d" is
Dimension of Embedding

Not
length of input Sequence

